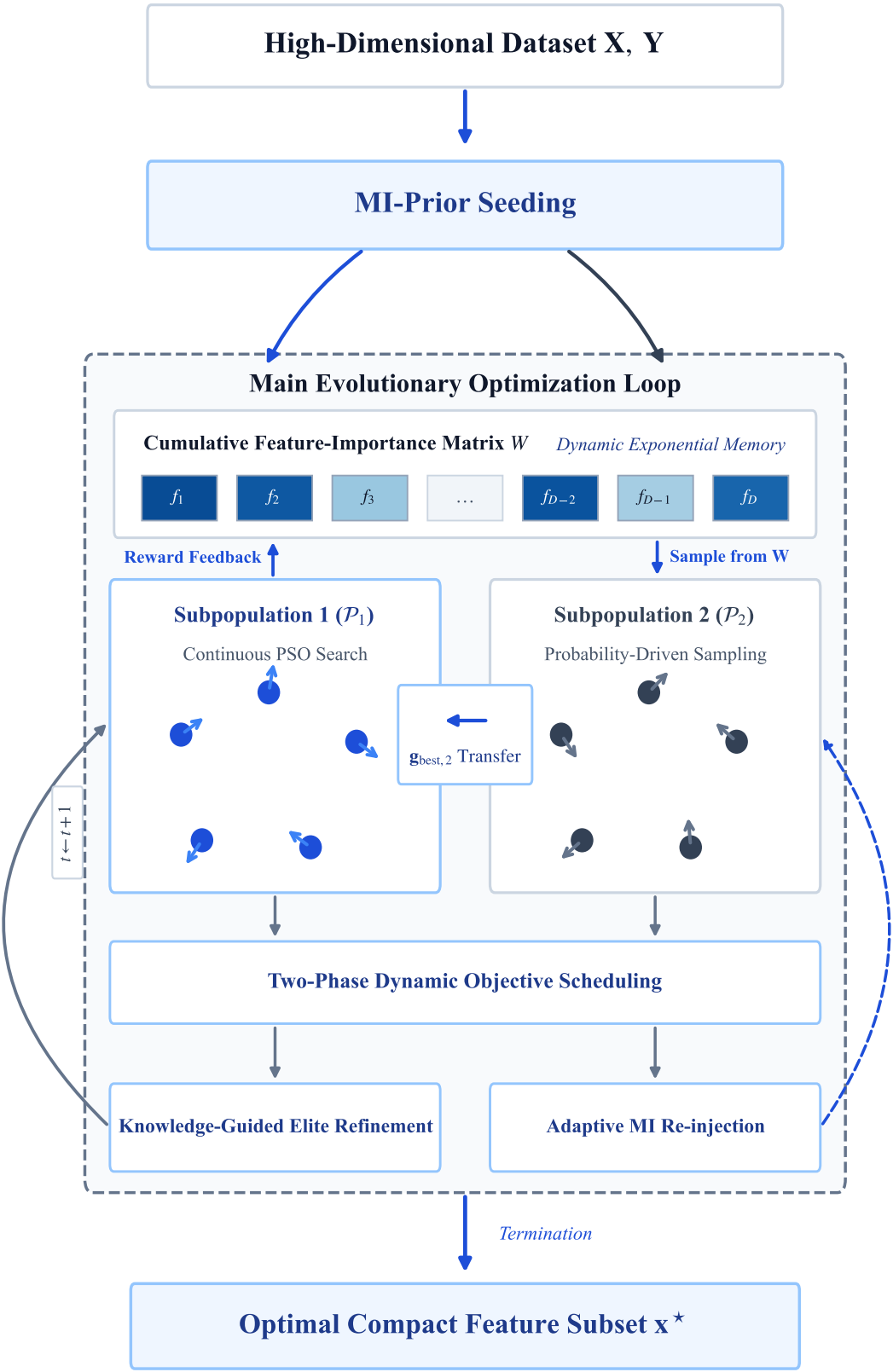

High-Dimensional Dataset X, Y
MI-Prior Seeding
Main Evolutionary Optimization Loop
Cumulative Feature-Importance Matrix W
Dynamic Exponential Memory
f1
f2
f3
…
fD−2
fD−1
fD
Reward Feedback
Sample from W
Subpopulation 1 (P1)
Continuous PSO Search
Subpopulation 2 (P2)
Probability-Driven Sampling
gbest, 2 Transfer
t ← t + 1
Two-Phase Dynamic Objective Scheduling
Knowledge-Guided Elite Refinement
Adaptive MI Re-injection
Termination
Optimal Compact Feature Subset x⋆